

A Comparative Analysis of Bottom-Up and Top-Down Visual Attention

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Abstract

The human visual system employs two primary mechanisms for attentional guidance: a rapid, stimulus-driven bottom-up process and a slower, goal-directed top-down process. This study provides an empirical comparison of these two mechanisms by analyzing eye-tracking data. Participants viewed images under a free-viewing (bottom-up) condition and a task-oriented (top-down) condition. Cumulative fixation heatmaps and individual scanpaths were generated to visualize gaze behavior. Furthermore, a computational model of saliency (Itti, Koch, & Niebur, 1998) was implemented to provide a theoretical baseline for bottom-up attention. Our results demonstrate a clear modulation of gaze patterns by top-down control, shifting focus from areas of high low-level saliency to regions of high task-relevance.

1 Introduction

Visual attention is the cognitive mechanism that enables the selection of pertinent information from complex visual environments. This process is largely governed by two distinct systems: a bottom-up, stimulus-driven system and a top-down, goal-directed system.

Bottom-up attention is a rapid and involuntary process, driven by low-level features of a visual scene. Elements with high intrinsic saliency, such as strong luminance contrast, unique colors, or specific orientations, automatically capture gaze. Conversely, top-down attention is a slower, voluntary process guided by an individual's goals, knowledge, and current task. It allows us to actively search for specific objects or features by modulating our gaze patterns to prioritize task-relevant information.

This study aims to empirically investigate and visualize the functional differences between these two attentional modes. By analyzing eye-tracking data, we seek to quantify how gaze allocation shifts when a viewing task transitions from unconstrained exploration (bottom-up) to goal-directed search (top-down). We further compare these human behavioral patterns to the predictions of a classic computational model of bottom-up saliency.

2 Methods

2.1 Experimental Design and Paradigms

To investigate the two primary modes of visual attention, we analyzed data from two distinct experimental paradigms: a task-oriented Top-Down condition and a free-viewing Bottom-Up condition. Crucial events, such as the onset of stimuli and user responses, were logged as timestamped messages within the data stream for precise temporal analysis.

2.2 Data Acquisition and Preprocessing

Gaze data was recorded using a high-frequency video-based eye tracker. The raw data, which included gaze coordinates, pupil size, and event markers, was stored in the EDF format. For analysis within MATLAB, these files were converted to '.mat' structures using the Edf2Mat toolbox, preserving all event messages. A final preprocessing step involved normalizing gaze coordinates from the native recording resolution (1920x1080) to the stimulus dimensions (1280x720).

2.3 Data Analysis Pipeline

The raw gaze data was segmented into discrete trials using the stimulus IDs and event messages (e.g., StartStimulus). Subsequent analysis focused on two primary visualization techniques.

2.3.1 Fixation Heatmap Generation

The generation of cumulative fixation heatmaps was performed through a systematic, multi-step process managed by our custom MATLAB functions. First, for each experimental condition (Bottom-Up and Top-Down), all relevant subject data files were loaded programmatically into a cell array.

For a given stimulus, the function 'collectFixationsForTrial' iterated through every subject's data. It used the event messages and timestamps to isolate the precise time window corresponding to the stimulus presentation. All fixation coordinates (x, y) occurring within this window were extracted. To ensure robustness against variations in the data structure, the extracted coordinate vectors were explicitly reshaped into a consistent columnar format. These individual fixation lists were then aggregated into a single $N \times 2$ matrix containing all fixations from all subjects for that specific trial.

Finally, the 'createAndSaveHeatmap' function took this aggregated matrix as input. It computed a 2D histogram to create a raw density map, which was

then normalized and smoothed using a Gaussian filter ($\sigma = 5$). The resulting heatmap was rendered as a semi-transparent layer ('alpha=0.5') over the original stimulus image and saved as a '.png' file.

2.3.2 Scanpath Visualization

To visualize the temporal dynamics and strategic patterns of eye movements, scanpaths were generated for a representative participant (Subject 1). This process was managed by our custom function, 'createAndSaveScanpath', which was designed for clarity and aesthetic quality.

The underlying data for the scanpath was gathered by the 'collectScanpathDataForTrial' function. For the specified subject and stimulus, this function first identified the precise temporal window of the trial using event markers. It then extracted two critical sets of information from this window: a time-ordered sequence of all fixation points, and the corresponding start and end coordinates of all interconnecting saccadic movements. These two lists were compiled into a single structured data object to be passed to the visualization function.

The 'createAndSaveScanpath' function then rendered this structured data onto the base stimulus image. Saccades were drawn as continuous, semi-transparent colored lines to trace the path of the eye's travel across the scene. The fixation points were overlaid on this path as solid, contrasting circular markers, indicating the locations where gaze paused. This visualization intentionally omits numerical labels or other distracting elements to provide a clean and intuitive representation of the subject's viewing strategy, allowing for direct qualitative comparison between the bottom-up and top-down conditions. The final visual output was saved as a '.png' file.

2.4 The Itti-Koch Computational Saliency Model

To establish a theoretical baseline for purely bottom-up attentional guidance, we employed the computational saliency model developed by Itti, Koch, and Niebur. This model is a biologically inspired framework designed to simulate the pre-attentive and stimulus-driven mechanisms of the early visual pathway in the primate brain. Its core function is to predict which locations in an image are most likely to attract attention based solely on their low-level physical properties, without any top-down or semantic influence.

The architecture of the model is hierarchical and operates in several distinct stages. First, the input image is decomposed into a set of fundamental feature channels. These include an intensity channel (derived from the average of the red, green, and blue color channels), two color-opponency channels (red-green and blue-yellow), and orientation channels. To compute the orientation information, the intensity map is filtered using Gabor filters at four different preferred orientations (0° , 45° , 90° , and 135°).

Each of these initial feature maps is then processed using a center-surround mechanism, which is inspired by the receptive fields of neurons in the retina and the lateral geniculate nucleus (LGN). This operation is implemented by computing the difference between a fine-scale version (the center) and a coarse-scale version (the surround) of each feature map. This is achieved by creating a Gaussian pyramid for each feature map and then calculating pixel-wise differences between maps at different scales. This center-surround operation is critical for highlighting locations that stand out from their immediate vicinity, such as a red dot in a green field or a horizontal line among vertical lines.

The outputs of the center-surround operations result in a set of feature maps. All feature maps corresponding to a single feature type (e.g., all color-opponency maps) are then normalized and summed together to create a conspicuity map. This results in three distinct conspicuity maps: one for intensity, one for color, and one for orientation. Each conspicuity map represents the overall saliency of the scene with respect to that one feature dimension.

In the final stage, these three conspicuity maps are themselves normalized and linearly combined into a single, master saliency map. This final map is a grayscale topographical representation of the visual scene, where the brightness of each pixel corresponds to its predicted visual saliency. The locations with the highest intensity in this map are the model's predictions for the most salient regions, which are expected to attract attention first in a bottom-up, free-viewing context.

3 Results

The analysis of the eye-tracking data revealed systematic and interpretable patterns of visual attention that varied with the experimental condition. This section presents a detailed, image-by-image analysis of the findings from a set of four unique visual stimuli.

For each stimulus, we present the empirical and computational results. The empirical findings consist of cumulative fixation heatmaps and representative scanpaths derived from the bottom-up and/or top-down paradigms in which the image was presented. The computational results consist of a predicted saliency map generated by the Itti-Koch model, which serves as a consistent baseline for bottom-up, stimulus-driven attention across all images.

The subsequent subsections will dissect the results for each unique image.

3.1 Analysis of Stimulus 1

The first stimulus was presented under both bottom-up and top-down conditions. For the top-down paradigm, participants were asked to determine if the family depicted was poor. This task-specific goal provides a clear framework for interpreting the resulting gaze patterns.

The cumulative fixation heatmaps are presented in Figure 1. The bottom-up heatmap (A) reveals that, without a specific task, attention was naturally drawn to the central cluster of human figures. This suggests an intrinsic attentional bias towards socially relevant elements. The top-down heatmap (B) shows a strikingly similar distribution of gaze. To answer the question about the family’s wealth, participants focused on the same key elements: the worn clothing of the man entering, the simple attire of the woman and children, and the sparse furnishings of the room. In this case, the visual cues for socio-economic status (top-down task)

were located in the same regions as the most naturally salient features (the people themselves).

This alignment between task-relevance and natural saliency is a significant finding. However, a divergence is noted when comparing human gaze to the computational model. The Itti-Koch saliency map (C) correctly highlights the human figures due to feature contrast but also assigns high saliency to the bright window on the left. This region was largely ignored by human observers in both conditions, indicating that semantic context (evaluating poverty) and social importance (faces) override low-level feature saliency.

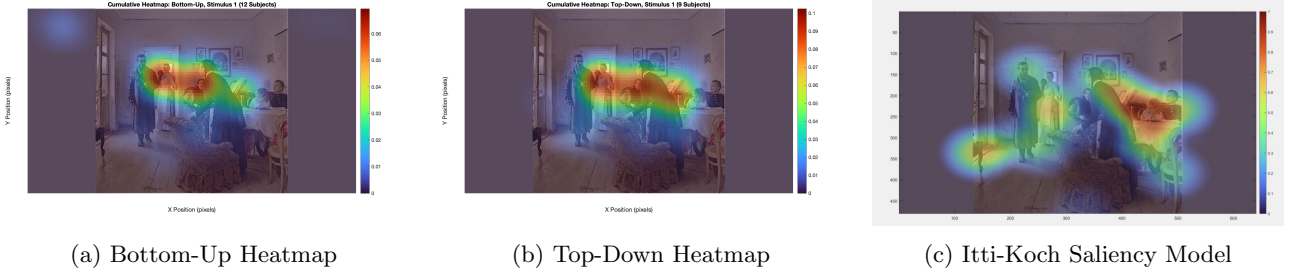


Figure 1: **Gaze Distribution for Stimulus 1.** A comparison of empirical heatmaps and the computational model. (a) The bottom-up heatmap shows natural focus on human figures. (b) The top-down heatmap displays a very similar pattern as participants assessed the family’s condition. (c) The Itti-Koch model prediction, which overemphasizes the non-semantic saliency of the window.

The scanpath analysis in Figure 2 further elucidates the viewing strategies. Both the bottom-up and top-down scanpaths show a dense pattern of saccades between the man and the woman, suggesting a focus on the scene’s central narrative. However, the top-down scanpath is subtly more purposeful. It includes

fixations on details relevant to the poverty question, such as the condition of the floor and the simple furniture, which are less prominent in the more exploratory bottom-up scanpath. This illustrates how a high-level cognitive task guides fine-grained visual interrogation of a scene.

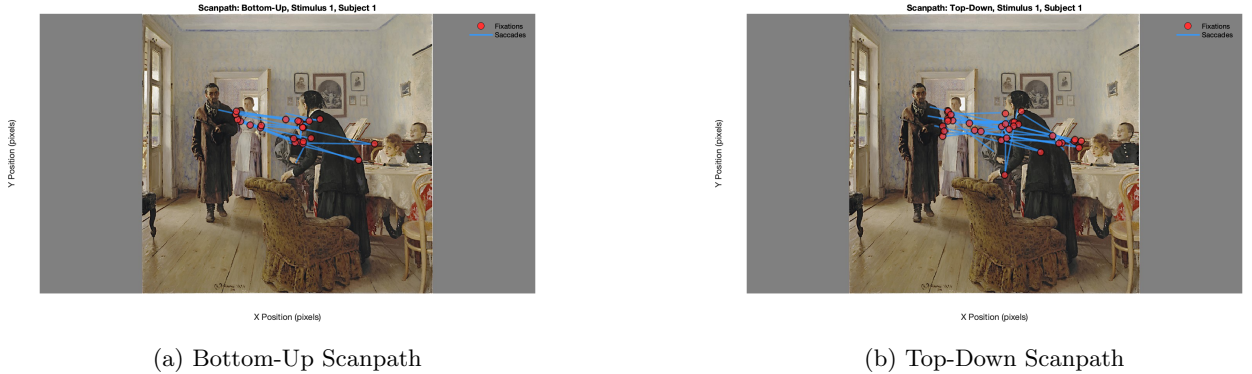


Figure 2: **Scanpath Comparison for Stimulus 1.** The viewing strategy of a representative participant under the (a) bottom-up and (b) top-down conditions. The top-down scanpath is more spatially concentrated and includes fixations on details relevant to the family’s economic status.

3.2 Analysis of Stimulus 2: The Time-Telling Task

The second stimulus was presented exclusively under the top-down condition, for which participants were asked to identify the time displayed on the clock. This highly constrained task provides a clear case study for goal directed attentional filtering.

The results, shown in Figure 3, demonstrate a powerful dissociation between human goal-directed attention and model-predicted saliency. The empirical heatmap from the top-down task (A) reveals an intensely focused concentration of gaze squarely on the clock face. The primary hotspot is centered on the clock hands, with secondary attention paid to the sur-

rounding numerals. This indicates that participants efficiently allocated their cognitive resources, ignoring all irrelevant visual information to perform the task.

In stark contrast, the Itti-Koch saliency model (B) predicted that the most salient region of the image was the cluster of colorful beads and textured material in

the upper right quadrant. The model, driven by low-level features, failed to identify the clock face as a region of interest. This highlights the model’s inability to process task relevance, a key function of the human top down attentional system.



(a) Top-Down Heatmap



(b) Itti-Koch Saliency

Figure 3: **Gaze Analysis for Stimulus 2.** (a) The top-down heatmap shows gaze is precisely locked onto the clock face to determine the time. (b) The computational model incorrectly predicts high saliency for the colorful beads, ignoring the task-relevant clock.

The scanpath for this task further illustrates this goal-directed strategy. The participant’s gaze lands on the clock and initiates a series of short, interrogative saccades around the clock’s center, followed by orienting saccades to the primary numerals. The scanpath is almost entirely confined to the clock, demonstrating a highly efficient filtering of salient-but-irrelevant distractors present in the rest of the scene.

3.3 Analysis of Stimulus 3: The Business Meeting

The third stimulus, depicting a group of people in a meeting, was presented under both experimental conditions. The top-down task required participants to determine if the discussion was serious, a high-level social inference task.

The resulting heatmaps, shown in Figure 4, illustrate a significant shift in viewing strategy. Under the bottom-up condition (A), attention is divided be-

tween the most prominent faces, resulting in two distinct hotspots over the central male speaker and the female listener to his right. This reflects a natural tendency to scan faces in a social scene. However, in the top-down condition (B), these separate hotspots merge into a single, larger area of interest. To assess the seriousness of the discussion, participants integrated visual information from the speaker’s face, his expressive hand gestures, and the documents on the table, focusing on the interaction space itself.

The Itti-Koch saliency model (C) provides a critical point of comparison. Its prediction bears a closer resemblance to the empirical bottom-up heatmap (A) than to the top-down one. The model correctly identifies the faces as generally salient due to low-level feature contrast, resulting in diffuse hotspots similar to those seen in the free-viewing condition. However, it fails to capture the focused, integrated pattern of the top-down task.



(a) Bottom-Up Heatmap



(b) Top-Down Heatmap

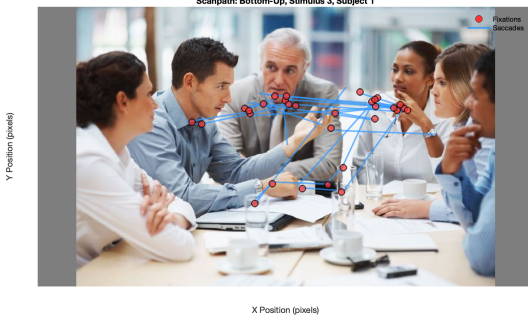


(c) Itti-Koch Saliency Model

Figure 4: **Gaze Distribution for Stimulus 3.** (a) The bottom-up heatmap shows distinct focus on individual faces. (b) The top-down heatmap reveals a unified area of attention covering the interaction space. (c) The computational model identifies faces but fails to capture the relational context.

This strategic shift is further detailed in the scanpath analysis (Figure 5). The bottom-up scanpath is characterized by saccades primarily jumping between the different faces. In contrast, the top-down scanpath forms a triangle of fixations, connecting the speaker’s face, his hands, and the faces of the listeners. This pat-

tern suggests a more complex cognitive process, where the participant is actively integrating multiple sources of social and contextual cues to make a high-level judgment, a behavior that cannot be predicted by low-level saliency alone.



(a) Bottom-Up Scanpath



(b) Top-Down Scanpath

Figure 5: **Scanpath Comparison for Stimulus 3.** The (a) bottom-up path shows face-to-face scanning, while the (b) top-down path integrates faces, hands, and context to infer the mood of the discussion.

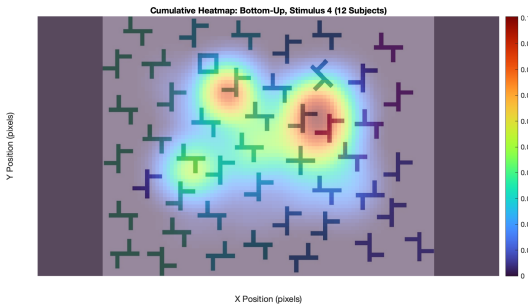
3.4 Analysis of Stimulus 4

The final stimulus, a visual array of abstract shapes, was presented only under the bottom-up, free-viewing condition. This allows for a direct comparison between human stimulus-driven attention and the predictions of the computational saliency model.

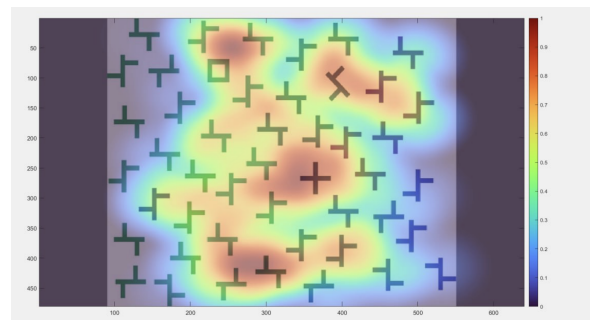
The empirical heatmap, shown in Figure 6 (a), indicates that human attention was not uniformly distributed. Instead, a few specific items captured a disproportionate amount of gaze. The most prominent hotspots are located on shapes that exhibit unique color properties, such as the reddish element in the

center-right and the brighter green element in the upper-left. This demonstrates the pop-out effect, where items with unique low-level features automatically attract attention.

The Itti-Koch saliency model’s prediction is presented in Figure 6 (b). The model’s output is conceptually aligned with the human bottom-up behavior, as it also identifies saliency based on feature contrast. It correctly highlights the same general regions that attracted human gaze. However, the model’s predicted saliency is far more diffuse and less decisive than the focused hotspots seen in the human data.



(a) Bottom-Up Heatmap



(b) Itti-Koch Saliency Model

Figure 6: **Gaze Analysis for Stimulus 4.** (a) The human bottom-up heatmap shows highly focused attention on a few pop-out targets. (b) The Itti-Koch model correctly identifies similar regions but produces a more diffuse and less selective saliency map.

The scanpath from the representative participant, overlaid on the stimulus, visually confirms this strategy. The gaze path consists of a series of long saccades that jump directly between the few most salient items, largely ignoring the surrounding distractors. This is a classic pattern of saliency-driven exploratory behavior,

where attention is involuntarily pulled from one pop-out location to the next.

4 Discussion

The results of this study provide a clear and compelling visualization of the interplay between bottom-up and top-down attentional systems. Across a diverse set of stimuli, we consistently observed that viewing behavior is a dynamic process, shaped by both the intrinsic properties of the visual scene and the internal goals of the observer.

In scenarios designed to elicit bottom-up attention,

human gaze patterns showed a strong tendency to favor regions with high low-level feature contrast, such as unique colors or prominent faces. This empirical finding aligns well with the theoretical predictions of the Itti-Koch saliency model. However, a key observation was that human attention is often more selective and focused than the model’s diffuse predictions, suggesting that even in the absence of an explicit task, higher-order factors (like the innate interest in faces) may contribute to attentional allocation.